ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CIT5241 INFORMATION RETRIEVAL AND WEB SEARCH

FINAL EXAMINATION

1st JUNE 2026

14:00 – 17:00

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Scenario:

Scenario: You are evaluating three resume screening systems (HireSmart, TalentScan, RecruitAI) for a recruitment agency. For each job posting, the system ranks candidate resumes. Each resume is manually labeled as Relevant (R) (shortlist for interview) or Not Relevant (N) (reject). The top-5 retrieved resumes are evaluated.

Data for Job Posting J1:

Relevance: R101 = R, R102 = R, R103 = N, R104 = N, R105 = N, R106 = R

Retrieved Results (Top-5):

HireSmart: [R101, R103, R105, R102, R104]

TalentScan: [R102, R101, R104, R103, R106]

RecruitAI: [R103, R105, R101, R104, R102]

Assume similar data exists for all 20 job postings.
Numerical mapping: Relevant (R) = 1, Not Relevant (N) = 0.

(a) For Job Posting J1 only, compute the following metrics for each resume screening system at retrieval depth 5:

i. Precision@5 – Show your calculations for all three systems. **[5 marks]**

ii. Recall@5 – Show your calculations for all three systems. **[5 marks]**

iii. Average Precision (AP) – Show step-by-step calculations for all three systems. **[8 marks]**

iv. Normalized Discounted Cumulative Gain (nDCG@5) – Use R=1, N=0. Show discounted gains, cumulative gains, ideal DCG, and final nDCG for all three systems. **[7 marks]**

(b) Evaluation across all job listings. Answer the questions below.

i. Using the AP values you computed for Job Posting J1 (above) and assuming the following AP values for the remaining 19 job postings shown in Table 1 below, compute MAP for each system. Show all working. **[12 marks]**

Job Posting	HireSmart (AP)	TalentScan (AP)	RecruitAI (AP)
J2	0.78	0.82	0.75
J3	0.85	0.88	0.83
J4	0.62	0.65	0.60
J5	0.91	0.89	0.92
J6	0.73	0.76	0.71
J7	0.68	0.71	0.66
J8	0.88	0.85	0.90
J9	0.55	0.58	0.54
J10	0.79	0.83	0.77

J11	0.94	0.92	0.95
J12	0.61	0.64	0.59
J13	0.82	0.86	0.80
J14	0.70	0.73	0.68
J15	0.87	0.84	0.89
J16	0.58	0.61	0.56
J17	0.76	0.80	0.74
J18	0.93	0.91	0.94
J19	0.64	0.67	0.62
J20	0.81	0.84	0.79

Table 1: AP values for the remaining 19 job listings for HireSmart, TalentScan and RecruitAI Systems

ii. Which system performs best based on MAP? Justify your answer with the computed values.
[5 marks]

iii. With binary relevance (R/N), nDCG@5 behaves identically to another metric you have computed. Which metric is it, and why? Would using nDCG@5 instead of MAP change your conclusion about the best-performing system? Explain.
[8 marks]
(Sub-Total: 50 marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

Test collection-based evaluation is a standard methodology in information retrieval (IR) to assess the effectiveness of retrieval systems. The evolution of test collections in information retrieval (IR) evaluation began with the Cranfield experiments in the 1950s and 1960s, which established foundational methods using fixed sets of documents, queries, and relevance judgments. This framework was significantly expanded by the Text REtrieval Conference (TREC) starting in 1992, which introduced large-scale, diverse collections and standardized evaluation protocols, facilitating broad IR research collaboration. Subsequent developments saw the creation of specialized collections for domains such as web, and multimedia.

a) Discuss with the aid of a suitable diagram, the process of generating relevance judgements for test collections **[15 Marks]**
b) Discuss the challenges that still exist for the test collection based evaluation methodology. Your discussion should also include attempts made by the research community in information retrieval to address these challenges. **[10 Marks]**

(Sub-Total: 25 marks)

Question 3

(a) List and briefly explain five distinct uses of machine learning in information retrieval systems. For each use, provide a one-sentence explanation of how machine learning improves that aspect of retrieval. **[10 marks]**

(b) Scenario: You are building a machine learning-based document retrieval system for a legal research platform. The system takes a user's search query (e.g., "contract breach cases") and classifies whether each document in a large legal database is relevant or not relevant to the query. After training your model, you evaluate it on a test set of 500 legal documents with known relevance judgments.
The confusion matrix below summarizes the performance of your binary classification model:

	Predicted Relevant	Predicted Not Relevant
Actual Relevant	180 (TP)	20 (FN)
Actual Not Relevant	50 (FP)	250 (TN)

Where:
TP (True Positive) = Relevant documents correctly classified as relevant
FN (False Negative) = Relevant documents incorrectly classified as not relevant
FP (False Positive) = Not relevant documents incorrectly classified as relevant
TN (True Negative) = Not relevant documents correctly classified as not relevant

Using the values above, compute the following evaluation metrics for your machine learning model. Show all formulas and step-by-step calculations.

i. Accuracy
[3 marks]

ii. Precision
[3 marks]

iii. Recall (also known as Sensitivity or True Positive Rate)
[3 marks]

iv. F-measure (also known as F1-score) – assume equal weight for precision and recall
[6 marks]

(Sub-Total: 25 marks)

Question 4: Vector Space Model for Movie Recommendation

(i) Describe, with the aid of an example, the procedure employed in the vector space model to generate rankings of movies given a particular user query (e.g., "action thriller with mystery plot").

[5 marks]

(ii) Using your answer in (i), generate a ranked list of movies given the query and the list of movie descriptions below. Show all working, including term frequency calculation, vector representation, and similarity computation.

Query: "action thriller mystery"

Movie Descriptions:

Movie A: "An action movie with thrilling car chases and mystery elements."

Movie B: "A mystery thriller that keeps you guessing until the end."

Movie C: "An action film focused on combat and explosions."

Movie D: "A thriller with action sequences and a mysterious antagonist."

[20 marks]

(Sub-Total: 25 marks)

END OF EXAMINATION